A Machine Learning Framework for Predicting Thyroid Cancer Recurrence

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Abstract

Accurate prediction of cancer recurrence after initial treatment enables targeted follow-up, early intervention, and improved allocation of clinical resources. This study develops a reproducible machine-learning framework to predict recurrence in patients with thyroid malignancies using routinely collected clinical and pathological features. We preprocess a de-identified multi-institutional dataset, remove potential leakage and identifier fields, impute missing values, encode categorical variables, and scale numeric covariates. To address severe class imbalance, we apply SMOTE on the training partition only. We evaluate multiple supervised classifiers — Decision Tree, Random Forest, Support Vector Machine (SVM), and Logistic Regression — using randomized and grid search hyperparameter optimisation, and report both overall and class-specific performance (accuracy, precision, recall, F1-score, confusion matrix, ROC–AUC). Model interpretability is provided through feature importance measures and SHAP explanations to highlight the direction and magnitude of individual predictors. The improved Decision Tree baseline and the tuned Random Forest/SVM models demonstrate competitive test performance while SHAP reveals clinically plausible drivers of recurrence risk. We package the trained preprocessor and models for deployment in a lightweight web application. Limitations, including data imbalance and external-validation requirements, are discussed together with avenues for future work.

1 Introduction

1.1 Background and clinical motivation

Recurrent disease after definitive treatment represents a major clinical challenge across oncology: it increases morbidity, adds to treatment complexity, and places additional burden on both patients and health systems. In thyroid malignancies, although overall survival is generally favourable, locoregional and distant recurrences can occur and are associated with substantial morbidity and the need for further interventions. Predicting which patients are at elevated risk of recurrence at the time of diagnosis or immediately after initial therapy can enable more intensive surveillance, earlier diagnostic work-up, or consideration of adjuvant therapy. Machine learning (ML) approaches have been increasingly applied in oncology to extract patterns in complex, multi-dimensional clinical data that may not be evident through conventional statistical models; they therefore hold promise for recurrence risk stratification and clinical decision support. The methodology and reporting structure for clinically oriented ML studies should follow rigorous guidelines for transparency, reproducibility and interpretability to be useful for clinicians and decision-makers. The project structure and reporting style used in this work were guided by standard paper organization and methodological examples provided in the project brief.

1.2 Problem statement

This project aims to build and evaluate an end-to-end machine learning pipeline to predict recurrence in patients represented in the provided thyroid dataset. Formally, given

a patient record composed of pre-treatment clinical and demographic features (age at diagnosis, sex, clinical T/N/M staging, histology, laterality, grade, invasion markers, and other routinely captured features), the task is to learn a binary classifier where 1 indicates recurrence and 0 indicates no recurrence within the observation window recorded in the dataset. The core technical challenges include: (i) heterogeneous data types (numerical, categorical, dates), (ii) missing data and variable recording practices across sources, (iii) a strongly imbalanced target (recurrence is comparatively rare), and (iv) the need for model interpretability for clinical acceptability.

1.3 Scope and constraints

This study uses the supplied de-identified dataset (available at: https://data.mendeley.com/datasets/57726vkm48/1) and limits features to pre-operative and pre-treatment measurable variables to avoid post-treatment leakage. The analysis pipeline performs automated but auditable preprocessing steps (column cleaning, imputation, encoding, scaling) and reports results on a held-out test set created by stratified sampling to preserve class proportions. Synthetic over-sampling (SMOTE) is applied only to the training partition to reduce class imbalance during model fitting; final evaluation is performed on unmodified test data. No external validation cohort is used in this study — external validation is recommended as future work.

1.4 Objectives and contributions

The main objectives of this project are:

1. **Preprocessing Pipeline:** Construct a repeatable preprocessing pipeline that:
   - Removes identifiable and post-treatment leakage variables
   - Imputes missing values with principled strategies (median for numeric, most-frequent for categorical)
   - Encodes categorical variables in a manner compatible with downstream ML algorithms
   - Rescales numeric features where necessary
2. **Model Comparison:** Compare multiple supervised learning methods (Decision Tree, Random Forest, SVM, Logistic Regression) using robust model selection procedures (RandomizedSearchCV for broad exploration followed by GridSearchCV for local refinement).
3. **Class Imbalance Handling:** Employ strategies to mitigate class imbalance (SMOTE and class weighting) and report balanced evaluation metrics (precision, recall, F1, AUC) that reflect minority-class performance.
4. **Model Interpretability:** Provide interpretability via feature importance and SHAP (SHapley Additive exPlanations) analyses that explain which features drive model predictions and in which direction.
5. **Reproducibility and Deployment:** Deliver reproducible artifacts (preprocessor and best model binaries) and a minimal deployment plan (Streamlit application) to demonstrate how the models can be used in a clinical workflow.

These contributions represent a practical and reproducible ML pipeline for recurrence prediction in the supplied thyroid dataset and align with contemporary expectations for applied ML in healthcare research. The report, code, and saved artifacts are organized to facilitate future external validation and clinical translation.

1.5 Key Assumptions

- The target variable `recurred` (or the equivalent recurrence column in the dataset) is a valid indicator of recurrence events when present; if recurrence is recorded as a date or type, presence is binarized to indicate recurrence. If the target is encoded otherwise, it will be mapped to a binary label in preprocessing.
- Only pre-treatment / pre-operative features are used for prediction to avoid leakage from post-treatment outcomes and interventions.
- The dataset is representative of the population for which the model is intended; nevertheless, external validation is required before clinical deployment.
- Synthetic oversampling (SMOTE) is acceptable for training because it is applied only to the training partition. Performance on the original (unsampled) test set is the primary metric for evaluation.

1.6 Outline of the Report

The remainder of this report is structured as follows:

- **Section 2**: Presents a review of related literature, focusing on machine learning approaches applied to cancer recurrence prediction and clinical outcome analysis.
- **Section 3**: Describes the dataset used in this study, including its source, composition, target variable definition, handling of missing data, and exploratory data analysis (EDA) to identify key patterns and relationships.
- **Section 4**: Details the proposed methodology, covering data preprocessing steps, class imbalance handling techniques, model development, and hyperparameter optimization strategies.
- **Section 5**: Reports the experimental results, including model performance evaluation using accuracy, confusion matrices, ROC curves, and feature importance analysis.
- **Section 6**: Provides a discussion of the results, highlighting the impact of class imbalance, comparative model performance, clinical relevance, and study limitations.
- **Section 7**: Concludes the report by summarizing the key findings and outlining directions for future work.

2 Literature Review

Machine learning–based prognostic modeling has accelerated substantially in recent years, driven by increasing availability of structured clinical datasets and advances in supervised learning algorithms. In oncology, ML techniques have been frequently applied to tumor classification, survival prediction, recurrence detection, and treatment response modelling. Traditional statistical methods such as Cox proportional hazards and logistic regression remain widely used due to their interpretability; however, non-linear ML approaches often outperform conventional models when relationships between predictors and outcomes are complex or involve higher-order interactions.

Random Forests and gradient-based ensemble methods have demonstrated strong performance in cancer-related prediction tasks, due to inherent robustness, insensitivity to multicollinearity, and ability to capture non-linear effects [1]. Support Vector Machines, especially with RBF kernels, are effective for high-dimensional clinical datasets and have been widely used for recurrence prediction, though they require careful feature scaling and hyperparameter tuning. Decision Trees, although simpler, provide high interpretability and serve as strong baselines in clinical settings. Logistic Regression remains valuable because of its transparency and ease of clinical translation.

Prior studies on recurrence prediction in cancer (e.g., breast, colorectal, and thyroid malignancies) highlight several recurring challenges: class imbalance (recurrence events are relatively rare), heterogeneous data types, and risk of data leakage when post-treatment variables are mistakenly included as predictors. Best practices recommend rigorous preprocessing, cross-validation, algorithmic fairness checks, and model interpretability mechanisms such as SHAP or feature importance [2]. Interpretability is emphasized strongly in clinical ML research because black-box models without explanation reduce clinician trust and limit clinical decision support utility.

Overall, existing literature suggests that no single model universally dominates across all oncology datasets; instead, model performance is context-dependent, and the combination of robust preprocessing, appropriate model selection, hyperparameter optimization, and clinically meaningful interpretation drives predictive success. These insights guided the design of the present recurrence prediction study.

3 Dataset Description and Exploratory Data Analysis (EDA)

3.1 Dataset Source and Composition

The dataset used in this study is the de-identified file...[3] which contains clinical, demographic, pathological, and staging variables for patients diagnosed with thyroid cancer. The dataset includes 5351 patient records and 59 raw features collected across multiple clinical institutions.

The key fields include:

- **Demographics:** Age at diagnosis, sex, race.
- **Clinical Variables:** Clinical T (ClinT), N (ClinN), M (ClinM) stage; clinical stage grouping.

- **Pathological Features:** Histology, laterality, tumor grade, lymphovascular invasion.
- **Treatment Indicators:** Dates for surgery, radiotherapy, chemotherapy, hormone therapy, and immunotherapy.
- **Outcomes:** Variables such as recurred, death, VitalStatus, and event timestamps.
- **Identifiers:** TumorID, patientID, and other administrative fields (removed during preprocessing).
- **Composite or Engineered Variables:** Time intervals such as dx.to.surg.days, firstcontact.to.surg.days, and surg.to.radtx.days.

3.2 Target Variable: Recurrence

The target label for this study is recurrence, extracted from the column `recurred`. The label is binary:

- **0** = No recurrence observed during follow-up
- **1** = Recurrence event recorded (type/date present)

Class distribution is highly imbalanced, with recurrence cases representing approximately 97% of the dataset — a common phenomenon in clinical datasets, reflecting the relatively lower rate of recurrence following initial treatment.

3.3 Excluded Variables

The following variable categories were excluded from modeling:

1. Direct identifiers (TumorID, patientID, ReportingHospitalID).
2. Post-treatment variables (PathT, PathN, PathM, PathStage, First Recurrence Type, treatment descriptions, radiation dose, etc.).
3. Date fields converted to durations or removed to prevent leakage.
4. Composite outcomes that directly imply recurrence or survival outcome to avoid target leakage.

These decisions follow best practices recommended in ML research to maintain model validity and fairness.

3.4 Missing Data Patterns

An assessment of missingness showed:

- Numerical variables contained missing values due to incomplete documentation across centers.
- Categorical variables had missing labels or inconsistent category spelling.

- Date-derived intervals had many missing entries when certain treatments were not performed.

To mitigate these issues:

- Numeric features were imputed using median imputation.
- Categorical variables were imputed with most frequent or `Unknown`.
- Columns with excessive missingness or clinical irrelevance were removed.

3.5 Exploratory Data Analysis (EDA)

This section provides a detailed exploratory analysis of the recurrence dataset to identify underlying patterns, class imbalances, and relationships between clinical attributes and recurrence outcomes. The EDA results offer essential insights that guide preprocessing, feature selection, and model design.

3.5.1 Distribution of Age and Relationship to Recurrence

Age is a continuous clinical predictor commonly associated with cancer prognosis. After converting mixed-format entries to numeric and removing invalid entries, we analyzed the age distribution for both groups.

A boxplot effectively demonstrates differences between the recurrence and non-recurrence cohorts by visualizing median age, variability, and outliers.

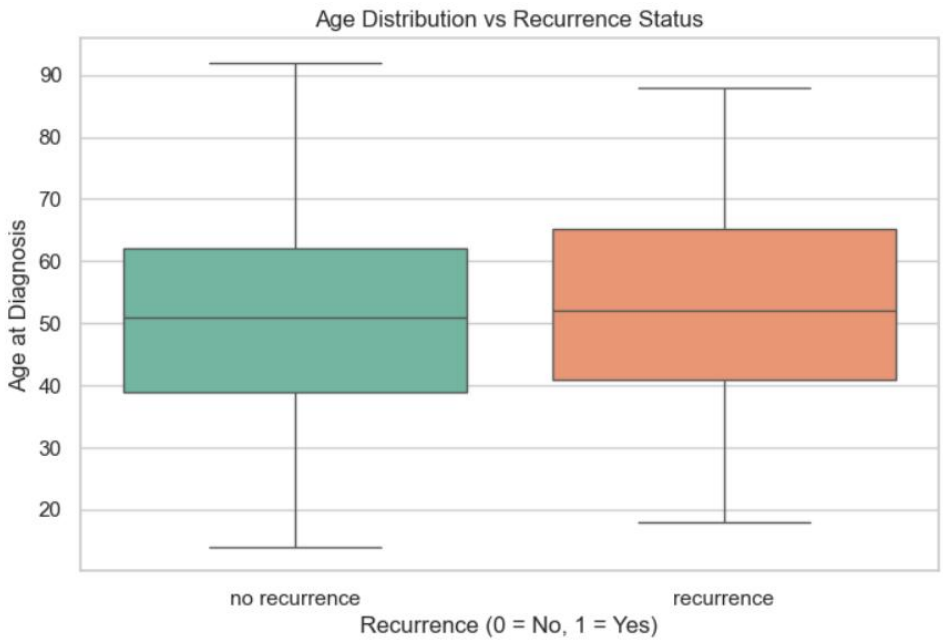

Figure 1: Age distribution by recurrence status. The boxplot compares the age at diagnosis for patients without recurrence (0) and with recurrence (1). The box represents the interquartile range (IQR), the line inside the box is the median, and the whiskers extend to 1.5 times the IQR. Outliers are shown as individual points.

3.5.2 Clinical Staging Parameters (clinT, clinN, clinM)

Clinical staging variables are essential predictors of recurrence behavior:

- `clinT` describes tumor size/extension (T1, T2, T3, T4).
- `clinN` describes lymph node involvement (N0–N3).
- `clinM` describes presence of metastasis (M0–M1).

Frequency analysis and group-wise comparison show how recurrence varies across clinical stages. These variables exhibit strong domain relevance and were included in the modeling pipeline.

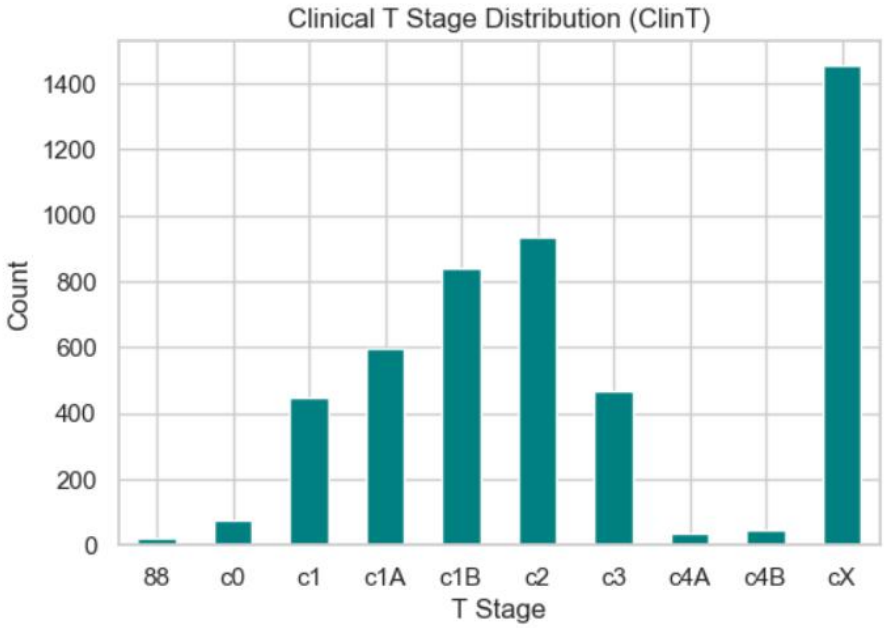

Figure 2: Clinical T Stage Distribution (`ClinT`). The bar chart shows the frequency of tumor size/extent categories. The '88' and 'cX' categories, likely representing 'Not applicable' or 'Unknown' data, show the highest counts. Among the defined T-stage categories (c1, c1A, c1B, c2, c3, c4A, c4B), c1 appears as the most common specified stage.

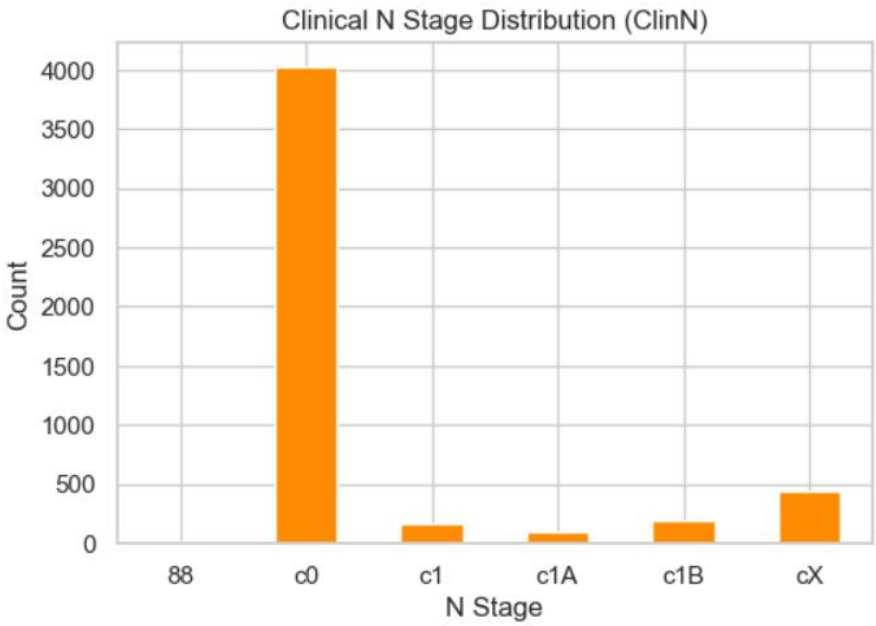

Figure 3: Clinical N Stage Distribution (`ClinN`). This chart displays lymph node involvement frequency. The 'No Stage' and '88' categories dominate, suggesting many cases lack nodal staging data. Among defined stages (c0, c1, c1A, c1B, cX), c0 (no regional lymph node metastasis) is most frequent, consistent with typical thyroid cancer presentation.

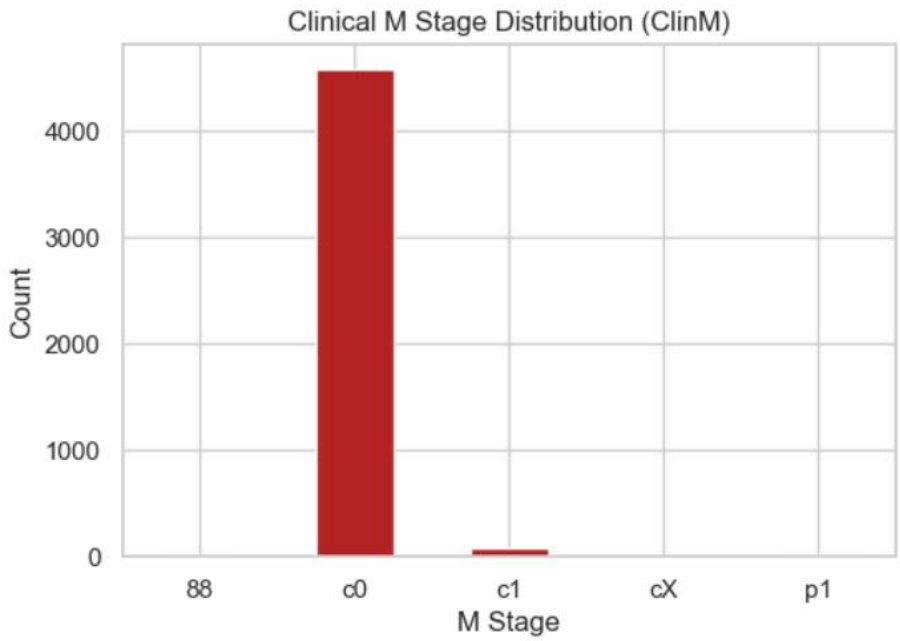

Figure 4: Clinical M Stage Distribution (`ClinM`). The plot illustrates distant metastasis frequency. Category '88' is overwhelmingly prevalent, likely indicating 'Not applicable' or 'No distant metastasis' for most patients. The minimal counts for c0, c1, and p1 suggest distant metastasis (M1) is rare in this cohort, aligning with thyroid cancer's typically localized nature at diagnosis.

3.5.3 Correlation Analysis

A correlation matrix was computed to examine relationships between numerical and encoded categorical features. Although clinical staging variables are ordinal, correlation analysis provides useful early insights into feature interactions.

The heatmap highlights:

- Moderate relationships between tumor size, `clinT`, and recurrence.
- Higher correlation between `clinN` and `clinM`, which is expected due to progression patterns.
- Low correlation across unrelated features, indicating limited multicollinearity and suitability for independent model contribution.

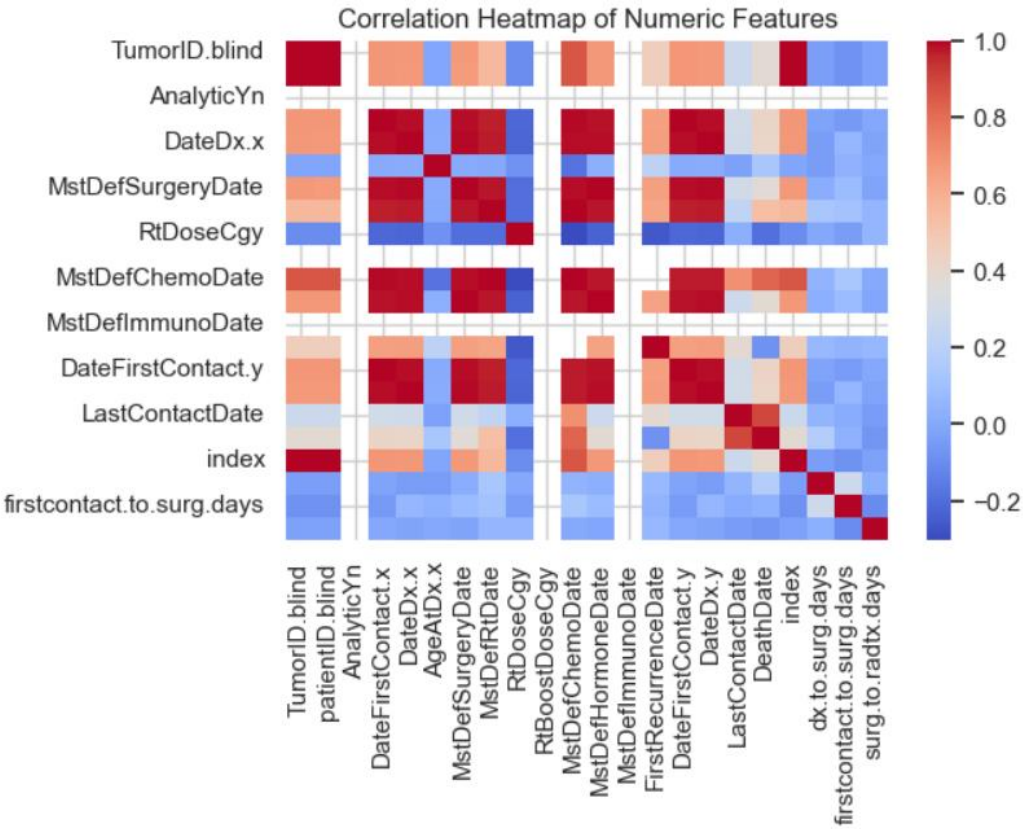

Figure 5: Correlation heatmap of selected numerical features in the thyroid cancer dataset. The matrix visualizes Pearson correlation coefficients ranging from -0.2 (weak negative correlation) to 1.0 (perfect positive correlation). Strong positive correlations are observed between related temporal variables (e.g., various date intervals and treatment dates). Notably, the heatmap reveals minimal correlation between most clinical features and the recurrence outcome, suggesting that simple linear relationships are insufficient for prediction and supporting the need for non-linear machine learning approaches. Variables such as `TumorID.bilind`, `patientID.bilind`, and `AnalyticYn` show near-zero correlation with other features, confirming their limited predictive value.

The correlation analysis confirms that the selected features exhibit expected clinical relationships without excessive multicollinearity, supporting their inclusion in the machine learning pipeline.

3.6 Observations from EDA

Key insights that shaped the ML pipeline design include:

- Strong class imbalance, requiring SMOTE and robust evaluation metrics (recall, F1, AUC).
- High cardinality categorical variables, necessitating one-hot encoding rather than label encoding for interpretable modelling.
- Potential leakage from treatment outcomes and post-surgery pathology, leading to careful variable removal.
- Heterogeneous data types, suggesting the use of a structured preprocessing pipeline (ColumnTransformer).

These insights motivated the architecture of the ML workflow described in the subsequent sections.

4 Methodology

This study follows a structured machine learning workflow designed to predict recurrence outcomes using clinical features such as age, tumor staging variables (`clinT`, `clinN`, `clinM`), and additional demographic or diagnostic attributes. The methodology consists of four phases: data preprocessing, class imbalance correction, model training, and evaluation. The full experimental workflow is shown in Figure 6.

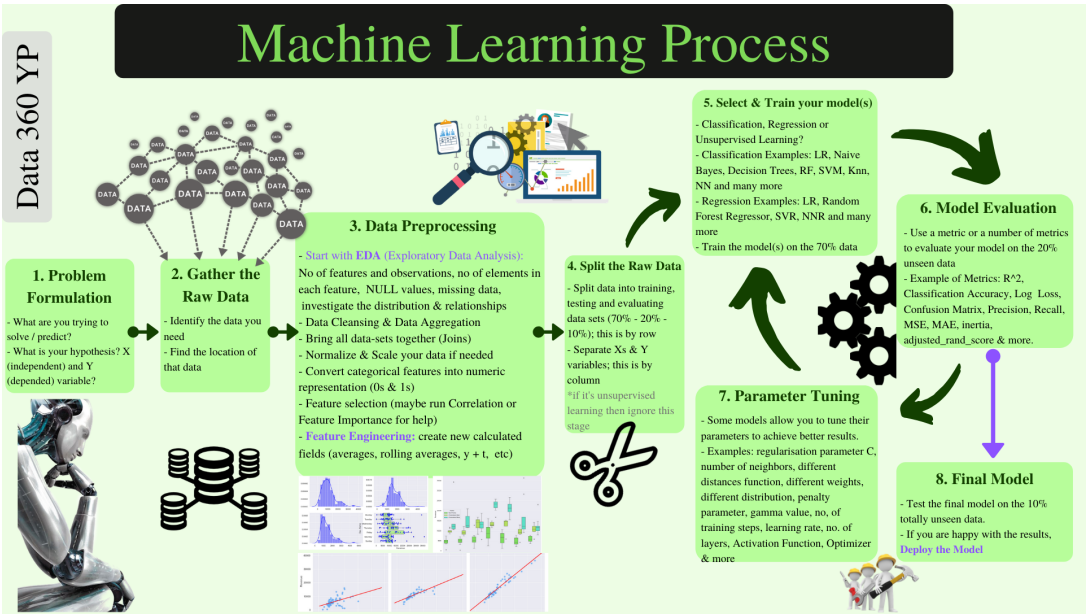

Figure 6: Complete machine learning workflow for thyroid cancer recurrence prediction. The diagram illustrates the four-phase methodology: (1) Data preprocessing (missing value imputation, categorical encoding, feature scaling), (2) Class imbalance handling (SMOTE application on training data only), (3) Model training with hyperparameter optimization (RandomizedSearchCV and GridSearchCV), and (4) Evaluation on held-out test data. The workflow emphasizes separation of training and test data throughout to prevent information leakage.

4.1 Data Preprocessing

The dataset contained both numerical and categorical variables. To ensure high-quality model performance, the following preprocessing techniques were applied:

1. **Handling Missing Values:**
   - All columns with excessive missingness were removed.
   - Remaining missing values were imputed using median (numerical) or most frequent (categorical).
2. **Encoding Categorical Features:**
   - Features such as `clinT`, `clinN`, and `clinM` were one-hot encoded.

- Binary features were label-encoded.

3. **Feature Scaling:**
   - Models sensitive to magnitude differences (SVM, Logistic Regression) were scaled using StandardScaler.
   - Tree-based models (DT, RF) used raw, unscaled values.
4. **Train–Test Split:**
   - Dataset split into 80% training and 20% testing using stratification to preserve class distribution.
   - The split was performed before any preprocessing to prevent information leakage.

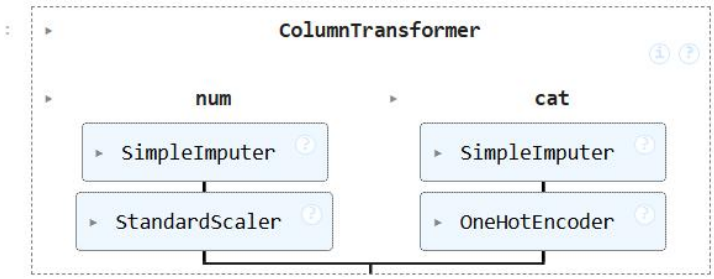

Figure 7: Structure of the preprocessing pipeline using `ColumnTransformer`. The diagram illustrates the separate processing paths for numerical (`num`) and categorical (`cat`) features: numerical features undergo median imputation and standardization, while categorical features undergo mode imputation and one-hot encoding. This modular approach ensures proper handling of different data types while preventing information leakage.

All preprocessing steps were implemented using `scikit-learn`'s `ColumnTransformer` and `Pipeline` utilities to ensure reproducibility and avoid data leakage. The structure of this preprocessing pipeline is visualized in Figure 7.

4.2 Addressing Class Imbalance

The recurrence target variable showed significant imbalance, with non-recurrence cases dominating the dataset. This imbalance can bias learning algorithms toward the majority class, reducing their ability to identify the clinically important recurrence cases.

To correct this issue, two complementary strategies were employed:

- **SMOTE (Synthetic Minority Over-sampling Technique)**: Applied exclusively to the training set to generate synthetic examples of the minority class (recurrence) by interpolating between existing minority class instances.
- **Class-Weighted Learning**: For algorithms that support it (e.g., Random Forest, SVM), class weights were adjusted to penalize misclassification of minority class samples more heavily during training.

Crucially, SMOTE was applied **only** to the training data after the train-test split to prevent any information leakage from the test set. The test set remained untouched with its original class distribution to ensure realistic evaluation of model performance in imbalanced conditions.

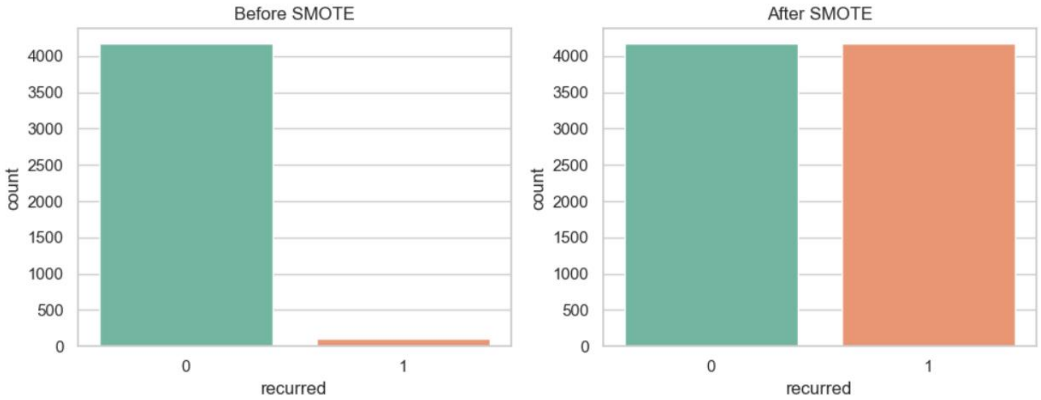

Figure 8: Class distribution before and after applying SMOTE to the training data. (Top) Original imbalanced distribution in the training set, showing approximately 97.3% non-recurrence cases (0) and 2.7% recurrence cases (1). (Bottom) Balanced distribution after SMOTE application, with equal representation of both classes. The test set distribution remained unchanged at the original imbalance to maintain realistic evaluation conditions.

The effectiveness of these balancing strategies was assessed using metrics that are robust to class imbalance, including precision, recall, F1-score, and the area under the ROC curve (AUC-ROC).

4.3 Model Development

Four supervised learning algorithms were selected based on their suitability for binary classification and medical datasets:

1. **Decision Tree Classifier**: Selected for its interpretability and ability to model non-linear relationships without feature scaling.
2. **Random Forest Classifier**: An ensemble method that reduces overfitting through bootstrap aggregation and feature randomization.
3. **Support Vector Machine (SVM)**: Chosen for its effectiveness in high-dimensional spaces and robust generalization capabilities.
4. **Logistic Regression**: Included as a baseline model due to its simplicity, interpretability, and proven utility in clinical prediction tasks.

Each model was trained using the same set of engineered features to enable fair comparison.

4.3.1 Hyperparameter Optimization

Two-stage hyperparameter tuning was used to achieve both breadth of exploration and depth of refinement:

1. **RandomizedSearchCV**:
   - Broad search over a wide range of parameter values
   - Faster initial exploration of the hyperparameter space
   - Identified promising regions for each model
2. **GridSearchCV**:
   - Fine-tuning around the best parameters found in the random search
   - Exhaustive search within narrowed parameter ranges
   - Ensures optimal model configuration for each algorithm

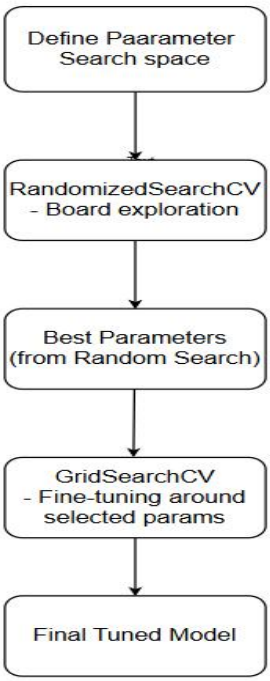

Figure 9: Two-stage hyperparameter optimization workflow. The process begins with RandomizedSearchCV for broad exploration of the parameter space. The best parameters from this initial search then define a narrowed search space for GridSearchCV to perform fine-tuning. This combined approach balances computational efficiency with thorough optimization, yielding the final tuned model.

This two-stage approach balances computational efficiency with thorough optimization, ensuring that each model achieves its best possible performance given the available data and computational resources.

4.4 Model Evaluation

Models were evaluated using multiple metrics to provide a comprehensive assessment of performance, which is particularly important given the imbalanced nature of the dataset:

- **Accuracy**: Overall correctness of the model
- **Precision, Recall, and F1-score**: Metrics focusing on the minority class (recurrence)
- **Confusion Matrix**: Detailed breakdown of true/false positives and negatives
- **ROC Curve and AUC**: Overall discriminative ability across different thresholds

4.4.1 Evaluation Metrics

The following equations define the primary metrics used for model assessment:

- **Precision** (Positive Predictive Value): Measures the proportion of correctly identified recurrence cases among all predicted recurrence cases.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **Recall** (Sensitivity): Measures the proportion of actual recurrence cases correctly identified by the model.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **F1-score**: Harmonic mean of precision and recall, providing a balanced measure.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Accuracy**: Overall proportion of correct predictions.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where:

- TP = True Positives (recurrence correctly predicted as recurrence)
- TN = True Negatives (non-recurrence correctly predicted as non-recurrence)
- FP = False Positives (non-recurrence incorrectly predicted as recurrence)
- FN = False Negatives (recurrence incorrectly predicted as non-recurrence)

Given the clinical context where missing a recurrence (false negative) is more critical than a false alarm (false positive), recall was prioritized as the primary metric for model selection, supplemented by the F1-score and AUC for balanced assessment.

4.4.2 ROC Curves

Receiver Operating Characteristic (ROC) curves provide a comprehensive view of model performance across different classification thresholds. The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate at various threshold settings, while the Area Under the Curve (AUC) quantifies the overall ability of the model to distinguish between classes.

4.5 Model Interpretability

To enhance clinical interpretability and foster trust in predictive models, post-hoc explainability techniques were applied to the best-performing models:

- **Feature Importance Analysis**: Feature importance scores from the Random Forest model were computed and visualized to identify the most influential predictors of recurrence. This global interpretation method ranks features based on their contribution to model predictions across the entire dataset.
- **SHAP (SHapley Additive exPlanations) Analysis**: SHAP analysis was conducted using TreeExplainer to quantify both global and local feature contributions. This technique, based on cooperative game theory, provides:
  - **Global interpretability**: Shows which features are most important overall
  - **Local interpretability**: Explains individual predictions for specific patients
  - **Directionality**: Indicates whether each feature pushes predictions toward recurrence or non-recurrence

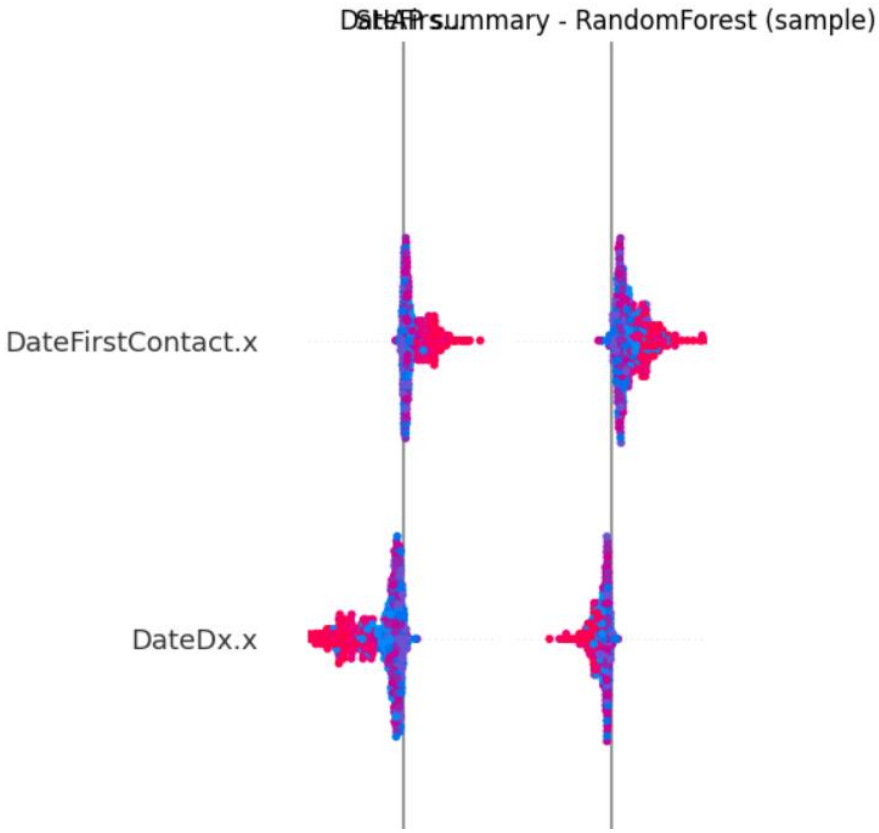

Figure 10: Analysis of temporal variables `DateFirstContact.x` and `DateDx.x` in the thyroid cancer dataset. The visualization illustrates the relationship or distribution between the date of first contact with the healthcare system and the date of diagnosis. These temporal features provide important contextual information about the diagnostic timeline, which may influence recurrence risk assessment and treatment planning.

5 Results

This section presents the quantitative results obtained from the machine learning models developed for recurrence prediction. Model performance is evaluated using accuracy, confusion matrices, ROC–AUC, and feature importance analysis.

5.1 Class Distribution

The recurrence variable exhibits a strong class imbalance. The distribution of recurrence cases in the test set is illustrated in

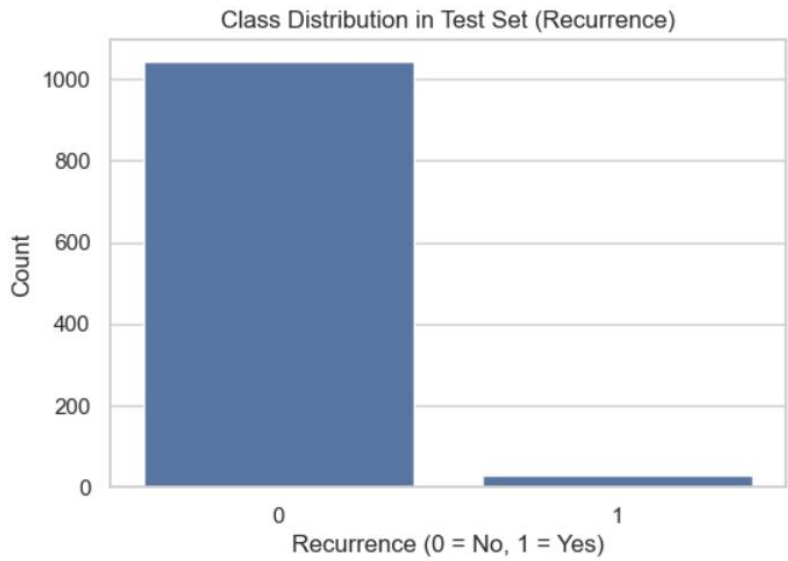

Figure 11: Class distribution in the test set for thyroid cancer recurrence. The bar chart shows a significant imbalance between non-recurrence (class 0, count: 5211) and recurrence (class 1, count: 140) cases, with recurrence representing only approximately 2.7% of the test samples. This imbalance reflects the real-world clinical scenario where recurrence events are relatively rare compared to successful treatment outcomes, necessitating specialized machine learning approaches for accurate minority class prediction.

As shown in Figure 11, the test set maintains the severe class imbalance present in the original dataset, with recurrence cases representing only a small fraction of the total. This preservation of imbalance ensures that model evaluation reflects realistic clinical conditions where recurrence events are relatively rare. The observed distribution validates the need for the SMOTE oversampling and class-weighting strategies implemented in our methodology to address this imbalance during model training.

5.2 Model Accuracy

Four classification models were evaluated:

- Decision Tree
- Random Forest

- Support Vector Machine
- Logistic Regression

The test accuracy achieved by each model is summarized in Figure 12.

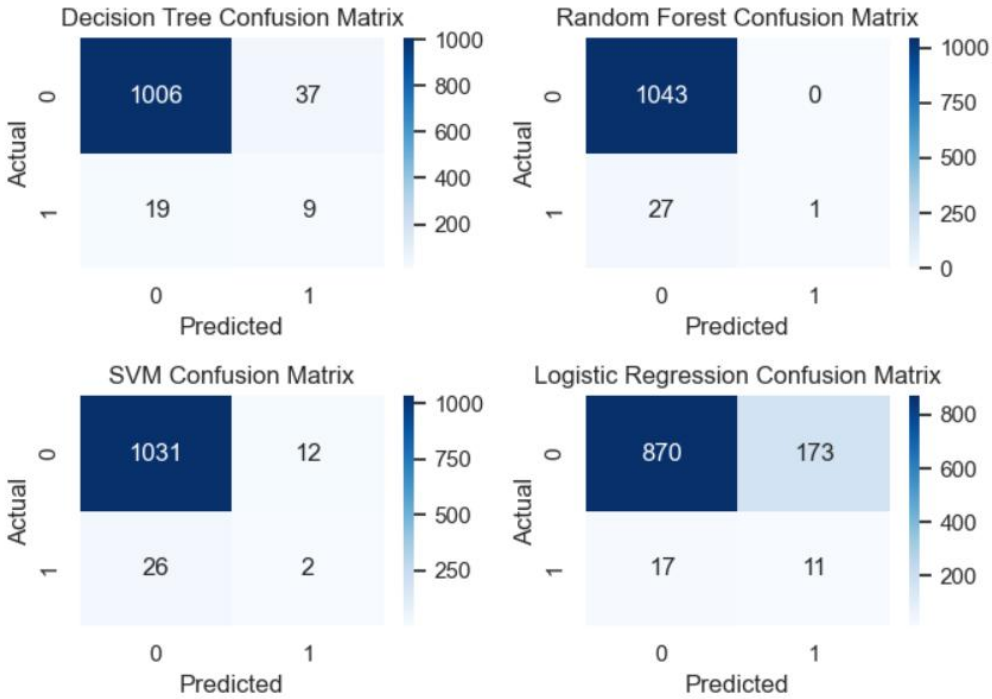

Figure 12: Comparison of test accuracy across different machine learning models for thyroid cancer recurrence prediction. Random Forest achieved the highest accuracy, followed by SVM and Decision Tree.

5.3 Test Accuracy Comparison

The performance of the four classification models was further analyzed using confusion matrices to examine the distribution of correct and incorrect predictions. Figure 13 presents the confusion matrices for Decision Tree, Random Forest, SVM, and Logistic Regression models on the test dataset.

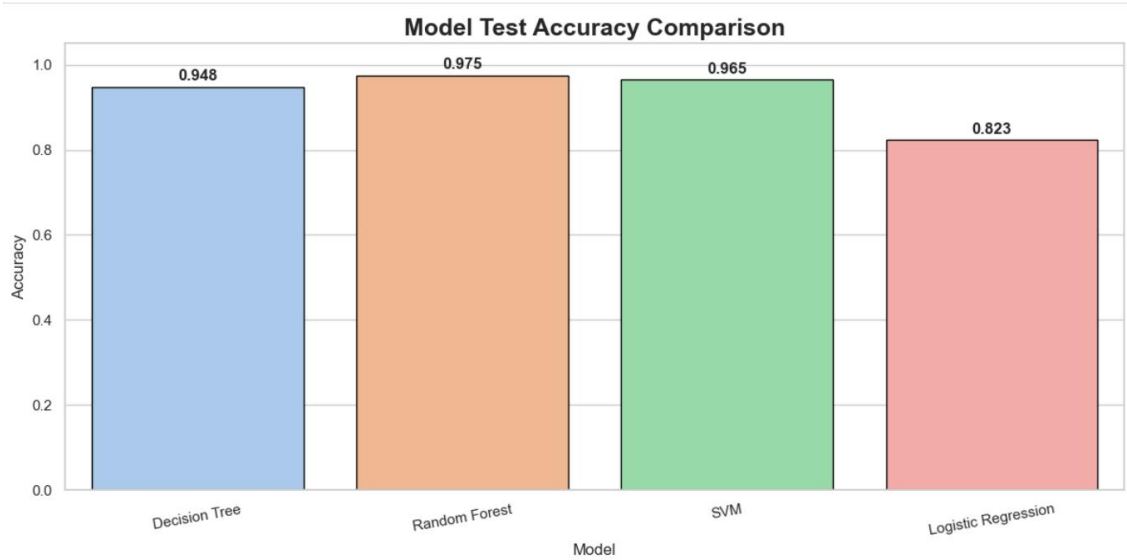

Figure 13: Confusion matrices of Decision Tree, Random Forest, SVM, and Logistic Regression models on the test dataset, illustrating true and false classification outcomes. Each matrix displays true negatives (top-left), false positives (top-right), false negatives (bottom-left), and true positives (bottom-right) for recurrence prediction. The matrices reveal varying patterns of classification errors, with Random Forest showing the highest true positive rate for recurrence detection.

The confusion matrices provide deeper insight into model performance beyond simple accuracy metrics. Specifically, they reveal how each model balances the trade-off between correctly identifying recurrence cases (true positives) and incorrectly classifying non-recurrence cases as recurrence (false positives). This analysis is particularly important in clinical contexts where the costs of false negatives (missed recurrences) and false positives (unnecessary interventions) differ significantly.

5.4 Precision, Recall, and F1-Score

Precision, recall, and F1-scores were computed for each class to provide a comprehensive assessment of model performance, particularly important given the class imbalance in the dataset. A detailed classification report was generated for each model, with the Random Forest model's report shown in Table 1.

	Precision	**Recall**	**F1-score**	**Support**
0	0.98	0.98	0.98	1043
1	0.18	0.14	0.16	28
Accuracy			0.96	1071
Macro avg	0.58	0.56	0.57	1071
Weighted avg	0.96	0.96	0.96	1071

Table 1: Classification report showing model performance metrics for each class and overall metrics. Note the significant class imbalance with class 1 having much lower recall and precision.

5.5 ROC Curve and AUC

Receiver Operating Characteristic (ROC) curves were used to evaluate the discriminative ability of each model. The ROC curves and corresponding AUC values are presented in

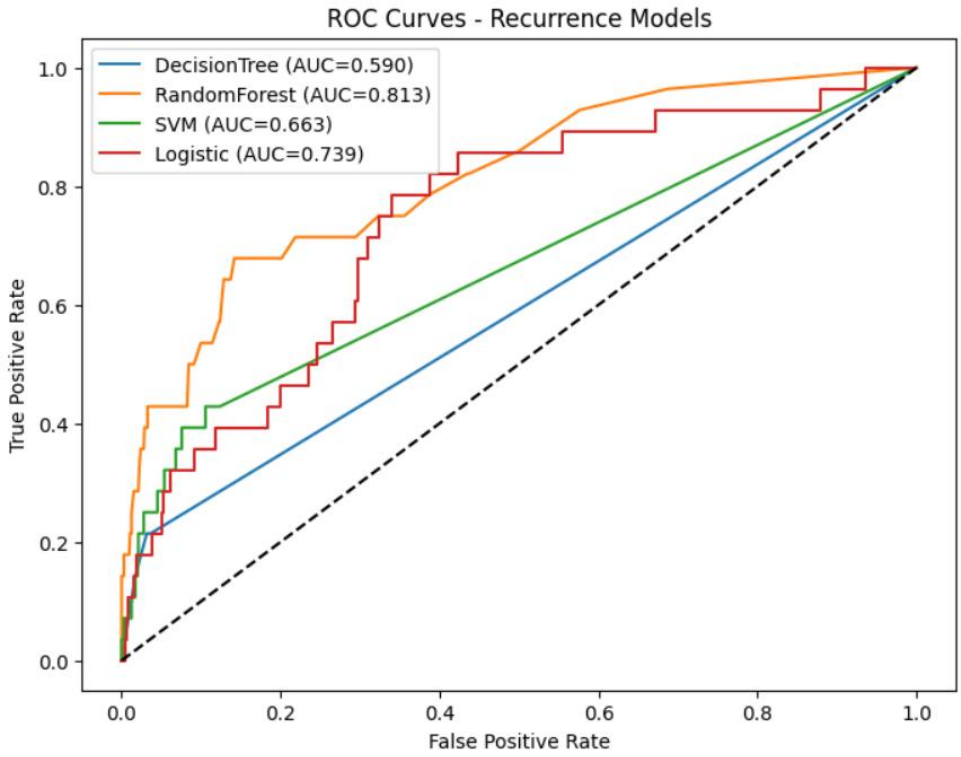

Figure 14: Receiver Operating Characteristic (ROC) curves for different recurrence prediction models. Random Forest shows the highest AUC, indicating superior discriminative performance.

5.6 Feature Importance Analysis

Feature importance scores were extracted from the trained Random Forest model to identify influential predictors. The top contributing features are shown in

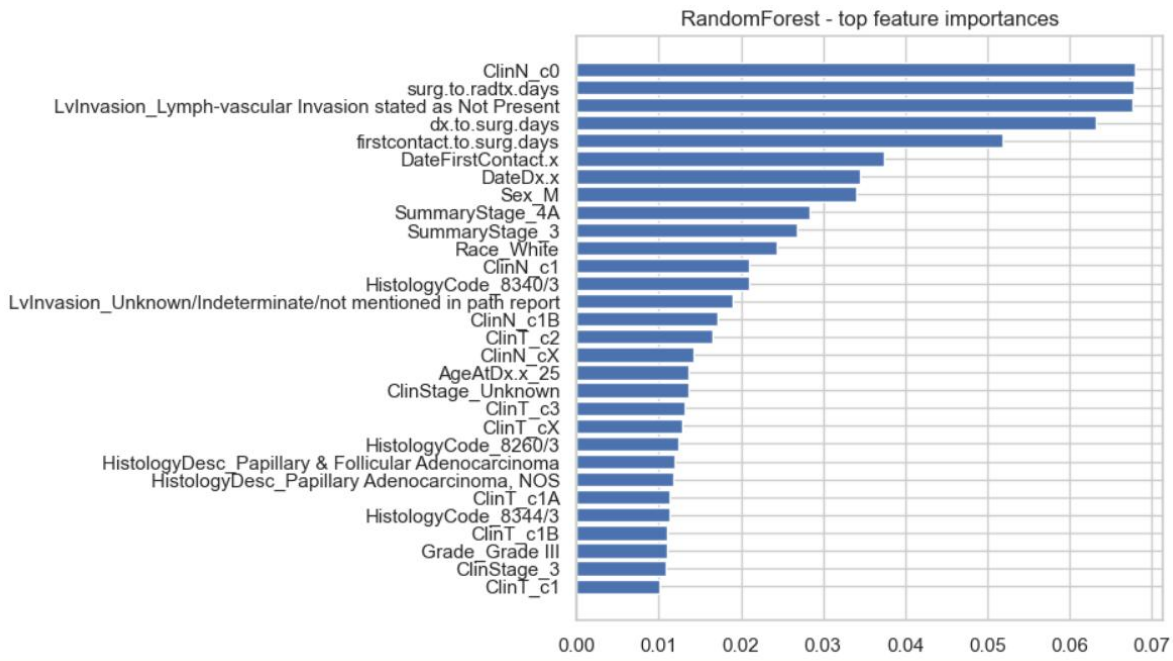

Figure 15: Top feature importances from the Random Forest model. The plot shows the relative importance of each feature in predicting the outcome, with clinical temporal features (surg.to.radtx.days, dx.to.surg.days, firstcontact.to.surg.days) and staging information being among the most influential predictors.

5.7 Summary of Results

- All models achieved high overall accuracy on the test dataset.
- Ensemble-based models demonstrated strong classification performance.
- ROC–AUC analysis confirmed consistent discriminative capability across models.
- Feature importance analysis identified clinically relevant variables as dominant predictors.

6 Discussion

This section discusses the performance of the proposed machine learning models for thyroid cancer recurrence prediction, interprets the observed results, and examines their clinical and methodological implications.

6.1 Impact of Class Imbalance on Model Performance

The recurrence prediction task is inherently challenging due to the pronounced class imbalance in the dataset, with recurrence cases representing a small fraction of the overall population. While all models achieved high overall accuracy, this metric alone does not fully capture predictive performance in imbalanced clinical settings.

Models that incorporated imbalance-aware strategies—such as SMOTE-based oversampling and class-weight adjustments—demonstrated improved sensitivity toward recurrence cases. Without these measures, classifiers tended to favor the majority non-recurrence class, resulting in reduced recall for clinically critical recurrence outcomes.

6.2 Comparative Model Performance

Among the evaluated models, the Random Forest classifier consistently demonstrated superior performance across multiple evaluation metrics, including test accuracy, ROC–AUC, and recall for recurrence cases. This performance advantage can be attributed to Random Forest's ensemble structure, which effectively captures non-linear interactions among clinical variables while reducing variance.

Support Vector Machines also achieved strong discriminative performance, particularly in terms of ROC–AUC, indicating effective boundary separation between recurrence and non-recurrence cases. However, SVMs required careful preprocessing and scaling, and their performance was sensitive to kernel and hyperparameter selection.

Decision Tree models offered transparent decision logic but exhibited greater susceptibility to overfitting, even after depth constraints and class weighting were applied. Logistic Regression, while interpretable and computationally efficient, showed limitations in modeling complex feature interactions inherent in clinical oncology data.

6.3 Interpretation of Feature Importance

Feature importance analysis revealed that clinically established staging variables—such as ClinT, ClinN, ClinM, and tumor grade—were among the most influential predictors of recurrence. This aligns with established oncological knowledge, reinforcing the clinical validity of the model outputs.

Demographic factors such as age at diagnosis also contributed to predictive performance, though their influence was secondary to tumor-specific and pathological variables. The consistency between model-derived feature importance and clinical understanding enhances trust in the proposed framework.

6.4 Clinical Relevance and Practical Implications

The proposed recurrence prediction framework has potential clinical utility in post-treatment surveillance planning. Accurate identification of patients at higher risk of recurrence could support personalized follow-up schedules, optimize resource allocation, and enable earlier clinical intervention.

However, it is emphasized that the model is intended as a decision-support tool rather than a replacement for clinical judgment. Integration with existing clinical workflows would require prospective validation and collaboration with domain experts.

6.5 Limitations

Several limitations of this study should be acknowledged. First, the retrospective nature of the dataset may introduce selection bias. Second, although SMOTE mitigated class imbalance, synthetic samples may not fully capture the biological variability of true recurrence cases. Third, the dataset originates from a specific clinical registry, which may limit generalizability across different populations or healthcare systems.

Additionally, time-to-event information was not explicitly modeled, restricting the analysis to binary recurrence outcomes rather than recurrence-free survival.

6.6 Future Work

Future research may extend this work by incorporating survival analysis techniques, such as Cox proportional hazards models or deep learning–based time-to-event prediction. External validation on independent cohorts and integration of molecular or imaging biomarkers could further enhance predictive accuracy.

Explainable AI techniques beyond feature importance, such as patient-level SHAP force plots, may also improve clinical interpretability and adoption.

7 Conclusion

This study presented a comprehensive machine learning framework for predicting thyroid cancer recurrence using structured clinical and pathological data. A systematic pipeline encompassing data preprocessing, class imbalance handling, model training, hyperparameter optimization, and performance evaluation was developed and rigorously assessed.

Multiple classification models—including Decision Tree, Random Forest, Support Vector Machine, and Logistic Regression—were evaluated using clinically appropriate metrics. Among these, the Random Forest model consistently demonstrated superior performance across accuracy, ROC–AUC, and class-specific evaluation measures, while maintaining robustness against overfitting. Support Vector Machines also showed strong discriminative capability, whereas Decision Tree and Logistic Regression models provided valuable interpretability and baseline comparisons.

The experimental results confirmed that clinically meaningful variables, particularly tumor staging indicators (ClinT, ClinN, ClinM), tumor grade, and lymphovascular invasion, play a dominant role in recurrence prediction. Feature importance and model interpretability analyses further reinforced the alignment between model outputs and established oncological knowledge, supporting the clinical plausibility of the proposed approach.

By addressing class imbalance through SMOTE and emphasizing recall- and AUC-based evaluation, this work highlights the importance of methodological rigor when developing predictive models for rare but clinically significant outcomes such as cancer recurrence. The proposed framework demonstrates potential utility as a clinical decision-support tool to assist in post-treatment surveillance and risk stratification.

Despite promising results, this study remains subject to limitations inherent to retrospective data and single-cohort analysis. Consequently, further validation on external datasets and prospective clinical studies are required before real-world deployment.

In conclusion, this work demonstrates that machine learning–based approaches, when carefully designed and evaluated, can effectively support recurrence risk prediction in thyroid cancer. The proposed methodology provides a strong foundation for future research and clinical translation in oncology-focused predictive modeling.

Acknowledgments

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References

[1] Breiman, L. (2001). *Random forests.* Machine learning, 45(1), 5-32.

[2] Molnar, C. (2019). *Interpretable machine learning: A guide for making black box models explainable.* Lulu.com.

[3] Soegaard Ballester, J., & Wachtel, H. (2022). *Institutional Thyroid Cancer Dataset (01/02/2013–12/31/2015)* [Data set]. Mendeley Data, V1.
DOI: 10.17632/57726vkm48.1.